

Trade Data Analysis

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Data Loading and Cleaning

STEPS FOR PREPARATION

The dataset was loaded using `pd.read_csv`. The **date** was converted to a **datetime format**, allowing for chronological sorting. After setting the date as the index, any **invalid dates** and **duplicate timestamps** were carefully removed to ensure data integrity.



Data Structure and Analysis

MISSING VALUES OVERVIEW

To ensure data integrity, we conducted a thorough analysis of the dataset, including:

- Utilization of `df.info()` for structure assessment
 - Identification of numeric and categorical columns
 - Examination of missing values in each column
 - Conversion of text-based numeric fields to appropriate types
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Statistical Insights and Trends

OVERVIEW OF TIME-SERIES ANALYSIS



Descriptive Statistics Overview

UNDERSTANDING DATA BEHAVIOR

- Utilized `df.describe()` to summarize key metrics
 - Observed mean, standard deviation, minimum, and maximum values
 - Analyzed distribution to reveal patterns
 - Gained insight into data scale and variability
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Daily and Cumulative Trends

ANALYZING TRADE FLUCTUATIONS

- Daily trade values plotted to **identify fluctuations**
 - Cumulative value plot reveals **long-term growth**
 - Peaks indicate high activity levels
 - Declines signal slowdowns in trade activity
 - These insights help in **making informed decisions**
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Analyzing Seasonal Trade Patterns

INSIGHTS ON TRADE VARIATIONS

- Weekly averages reveal peak trading days
 - Monthly analysis shows seasonal trends
 - Yearly comparisons highlight growth trajectories
 - Identified patterns assist in forecasting future trade activities
 - Seasonal fluctuations impact transport and logistics decisions
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Rolling Analysis and Volatility

UNCOVERING MARKET DYNAMICS

- Calculated a **30-day rolling mean** to smooth out fluctuations in trade values, revealing underlying trends.
 - Analyzed daily changes to identify significant jumps or drops.
 - Monitored rolling standard deviation to highlight periods of high volatility.
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Trade Attributes Insights

ANOMALY DETECTION METHODS



Country Contribution & Anomalies

INSIGHTS ON TRADE PATTERNS

- Analyze country-wise total trade values
 - Highlight major trading partners
 - Implement Z-score for anomaly detection
 - Identify unusual spikes or dips
 - Detect special events or data quality issues
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Final Output and Dataset Summary

CLEANED DATA READY

- Cleaned dataset saved as **clean_trade_detailed.csv**
 - Dataset prepared for reporting and dashboards
 - Insights from **20+ analyses** completed
 - Multiple time-series visualizations generated
 - Ready for further modeling and exploration
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Thank You for Your
Attention